K MOHITH KANNAN

B.Tech CSE (AI & ML) @ SRM Chennai | Python · Java · C · TypeScript | always-on systems, measured and tested

B.Tech CSE (AI & ML) · SRM Institute of Science and Technology · 2025–2029
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SUMMARY

Second-year CSE (AI & ML) student who builds software that keeps running when I am not watching it, and then measures it. A daily agent that finds and prepares my job applications, a second brain I talk to from my phone, and a video looper that lands within about 12 ms where competing ones overshoot by up to 250 ms.

PROJECTS

NitroWatch — a permission system for AI agents (governance layer for MCP) *Aug 2026 · NitroStack × SRM Hackathon*

- Connect an agent to an MCP server today and it is all-or-nothing: every tool, or none. NitroWatch sits in front of any MCP server, sorts each tool into three risk tiers — reads run instantly, reversible actions need approval, irreversible ones always do — lets safe tools earn autonomy through a clean track record, and writes a permanent audit trail.
- Built in a one-day campus hackathon on the NitroStack TypeScript SDK, and shipped with a public walkthrough documenting the six bugs found while building it — **two of them in the framework itself, reported upstream**.

TypeScript · MCP · policy engines · audit logging · github.com/Mohith535/nitrowatch

nova-cortex — an agent memory that is allowed to say “no” *Jul–Aug 2026*

- Every agent memory tool today is a bucket: add() always succeeds, so an agent can quietly store two contradictory facts and act on whichever it reads first. This one refuses the contradicting write, names the memory it conflicts with, and retires superseded facts instead of deleting them. The principle: **the model perceives, deterministic code governs, the database enforces** — the language model is never the authorization boundary.
- Exposed as an MCP server, so any MCP client can call it and be refused. Runs always-on (AWS Lambda + Bedrock, CockroachDB under serializable isolation, with no DELETE granted to the agent).
- I benchmarked my own thesis on a frozen, hash-locked corpus and it was refuted — the LLM out-governed my hand-written rules. I published that rather than bury it, then built selective escalation: roughly 92% of the model’s accuracy at a quarter of its calls. 190 tests; an adversarial agent I wrote found a real hole in my own soundness claim; seven prompt-injection attacks resisted.
- Its API is a FastAPI service: GET /health, POST /remember, GET /history, POST /recall, POST /surface.

Python · MCP · AWS Bedrock/Lambda · CockroachDB · benchmarking & adversarial testing

Nova — a ten-agent coaching layer over my own task engine *Jun–Jul 2026 · Google × Kaggle AI Agents Capstone*

- Ten specialist agents (Google ADK) reaching my data through **17 typed MCP tools**, built least-privilege on purpose: the Coach agent physically cannot write tasks, and the Planner has to get my confirmation before it commits anything.
- Grounded in real behaviour rather than self-report — it cites what I actually did (“you have postponed this three times”) instead of inventing encouragement.
- Its API is a FastAPI service: GET /, GET /api/context, GET /api/opportunities, GET /api/memory, DELETE /api/memory, POST /api/reset.

Python · Google ADK · Gemini · MCP · multi-agent orchestration · github.com/Mohith535/Nova-agentic-layer-over-taskflow

LoopLab — send someone an exact moment of a video (live product) *Aug 2026 · looplab.page*

- A dozen extensions loop a section of a YouTube video; not one lets you send it. LoopLab’s link opens for a stranger on any device with nothing installed, carrying a written note and a voice message. Framework-free TypeScript on Cloudflare Pages, so one core runs in both the web app and a browser extension.
- Competing loopers overshoot the end point by up to 250 ms; mine measures its own timing error every lap and lands within about 12 ms. 108 commits, 214 tests, live and in use.

TypeScript · Cloudflare Workers/Pages · Web Audio · edge functions

Continued on page 2 → TaskFlow: a task manager built on behavioural science, not features, and 2 more

PROJECTS — CONTINUED

TaskFlow — a task manager built on behavioural science, not features *Dec 2025 – present · v9.1.0*

- An execution engine rather than a list: one prime target a day because willpower depletes, deadlines that create urgency without anxiety, and one rule I hold every screen to — rescue, not punishment. If a screen makes you feel judged, it is wrong even when it is factually right.
- Fully offline by design: no telemetry, no accounts, local storage only. Every feature traces to published research (Gollwitzer's implementation intentions, the Zeigarnik effect, Yerkes–Dodson).
- Its web dashboard runs on a JSON REST API I wrote on Python's standard-library http.server (ThreadingHTTPServer), with GET and POST endpoints such as /api/tasks and /api/path, no framework.

Python · CLI · local-first architecture · atomic writes & crash recovery · github.com/Mohith535/TaskFlow

EDI — a second brain I talk to from my phone *Sep 2026 – present · in daily use*

- Phone → Telegram → Cloudflare Worker → queue → my laptop, so nothing is lost while the laptop sleeps. The worker holds no credential beyond Telegram and its speech-to-text key: fully compromised, it exposes one bot and one free key.
- Private entries are sealed with a public key before they touch storage, so the server can write them and can never read them back. 165 commits since September 2026.
- Built a second-brain system on a Telegram bot and a Cloudflare Worker with an evidence-gated memory model and confidence decay, so a stored fact has to keep earning its place rather than persisting unchallenged.

TypeScript · Cloudflare Workers + KV queue · Python · Telegram · speech-to-text

OPHunter — a daily agent that finds work for me and prepares the application *Jun 2026 – present · in daily use*

- Scans 12 job, hackathon and research sources every morning on GitHub Actions, ranks them against a target I set, and sends ready application packs to my phone through a Telegram bot on Cloudflare Workers, so it keeps working while my laptop is shut.
- A 10 s timeout was not stopping two career-board downloads that ran 236 s and 105 s, because the timeout only bounds the gap between bytes. I put each download under its own wall-clock deadline and fetched them in parallel: **218 s → 69 s**.
- Running it on live data exposed a bug I had written: listings stored only their first six courses, alphabetically, which cut "btech" off almost every list and hid 33 real internships from me as "not open to B.Tech". Fixed, with a test that fails on the old code.
- Replaced typed bot commands with a Telegram Mini App: switches, chips and one-tap buttons, signed in by checking Telegram's HMAC signature on every launch. The bot and my laptop share one setting through a token-protected GET/PUT REST API that answers 401 without the token and 409 when an older copy would overwrite a newer one. 400+ automated checks.

Python · GitHub Actions · Cloudflare Workers + KV · Telegram Mini App · REST APIs · github.com/Mohith535/opportunity-hunter

CAMPUS & COMMUNITY

- **Four hackathons in 2026:** NitroStack × SRM (Aug) · Hybrid Hack at Aaruush '26 (Jul) · Ossome Hacks 3.0 (Apr) · Bharath Hackathon, where I led a four-to-five member AI team on idea refinement, task allocation and technical direction.
- **Campus events — the technical ones:** a founders' meetup, an AR/VR workshop, and NVIDIA-led sessions at SRM. I go where something is actually being built or explained, and I usually leave with a question I then go and answer in code.
- **Writing and explaining:** technical posts on LinkedIn averaging 300–600 impressions, and public repositories written with real READMEs, architecture diagrams and an honest "what does not work yet" section — the part most students skip.

PROGRAMS & SELECTIONS

- **GCI World — Data Science Program**, Matsuo-Iwasawa Lab, University of Tokyo — completed (full pipeline: market analysis, EDA, modelling, evaluation and business proposal), 2026.
- **Google × Kaggle AI Agents Intensive** — five-day course and capstone; submitted a three-repository agentic system, 2026.
- **GSSoC 2026** (AI/Agents track) and **SSoC Season 5** — selected open-source contributor, 2026.
- **Deloitte Digital Camp 2026, Global Challenge** — designed intent-aware, policy-controlled AI systems.

TECHNICAL SKILLS

Languages: Python (advanced) · TypeScript · Java · C · C++ · JavaScript · SQL · HTML/CSS

AI & agents: AI coding agents (daily driver) · Model Context Protocol — servers and clients · Google ADK · Gemini · AWS Bedrock · multi-agent orchestration · prompt engineering · LLM benchmarking and evaluation · agent governance and safety

Systems: AWS Lambda · Cloudflare Workers/Pages · CockroachDB · PostgreSQL · key–value stores · REST APIs · Git and GitHub · CI · local-first and offline architecture

Practice: behavioural-science-driven UX · adversarial and property-based testing · privacy by design · technical writing and documentation

EDUCATION

- **SRM Institute of Science and Technology (SRMIST)** — B.Tech, Computer Science Engineering (Artificial Intelligence & Machine Learning) · 2025–2029
- **Sainik School Amaravathinagar** — Senior Secondary (PCM) · 2017–2024 · Ranked 2nd in class

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Also solved all 24 stars of Advent of Code 2025 in Python.